

A Quantity Theory of Money Approach to Cryptocurrency Valuation and Strategic Positioning

1 The Foundation: The Quantity Theory of Money ($MV = PQ$)

The valuation of a cryptocurrency, like any asset acting as money, can be traced back to the fundamental principles of monetary economics, specifically the **Quantity Theory of Money (QTM)**. The Fisher equation, $MV = PQ$, relates the total supply of money (M) and the rate at which it is spent (V , Velocity) to the nominal value of economic transactions (P , price level $\times Q$, real output).

In traditional fiat systems, this equation helps link money supply to inflation and economic activity. A critical insight is that for a digital asset, its market capitalization (MC) represents its money supply (M), and the economic output (PQ) it supports is the total value of transactions and services facilitated by the network. Therefore, $MC \approx \frac{PQ}{V}$, meaning an asset's value is fundamentally linked to the economic activity it processes, and the efficiency (velocity) with which it is used.

1.1 Decomposing Market Cap: The Role of λ and V_{Free}

A crypto asset's market cap (MC) is a summation of its **Store of Value (MC_{SoV})** and **Medium of Exchange (MC_{MoE})** functions. To formalize the split ($MC = MC_{\text{SoV}} + MC_{\text{MoE}}$), we decompose the circulating supply (M) based on its usage, using the **Total Value Locked (TVL)** as a guide.

The total token supply is partitioned into two components:

1. **Locked/Staked Supply (M_{Locked}):** Assets tied up in decentralized finance (DeFi), governance, or security protocols. These assets exhibit near-zero velocity.
2. **Free Circulating Supply (M_{Free}):** Assets actively used for transactions and trading.

We define λ (lambda) as the proportion of the total token supply that is **locked/staked**:

$$\lambda = \frac{M_{\text{Locked}}}{M}$$

The residual proportion, $(1 - \lambda)$, is the free circulating supply available for active use.

The market cap required to support the transactional volume (PQ) is determined by the velocity of the actively circulating supply. We define the **Free Velocity (V_{Free})** as the theoretical velocity if *only* the free supply (M_{Free}) was used for all transactions. V_{Free} represents the pure exchange efficiency of the actively circulating supply.

This V_{Free} is the critical input to value the MoE component using the QTM framework:

$$MC_{\text{MoE}} = \frac{PQ}{V_{\text{Free}}}$$

The remaining market capitalization is then attributed to the SoV function—reflecting long-term holding, security premium, and speculative potential:

$$MC_{SoV} = MC - MC_{MoE}$$

In this model, the MoE component is generated by the λ and V_{Free} factors, while the SoV component is the residual value attributed to the asset's function as a secure, long-term asset.

2 Strategic Metrics: NVT/G and the SoV/MoE Ratio

Based on this decomposition, we introduce two metrics to drive valuation and strategic positioning:

1. **NVT to Growth Ratio (NVT/G):** The Network Value to Transactions (NVT) ratio ($MC/\text{Annualized Transactions}$) is analogous to the P/E ratio for stocks. It assesses the market value relative to the asset's economic utility flow. We normalize it by dividing by the network's growth rate (G) (e.g., user growth, TVL growth) to create the NVT/G ratio, identifying whether the **MoE component** is priced efficiently relative to its adoption trajectory. A lower NVT/G suggests the MoE value proposition is more favorable.
2. **SoV/MoE Ratio:** This proprietary ratio ($\frac{MC_{SoV}}{MC_{MoE}}$) quantifies the asset's strategic balance. A high ratio indicates that the majority of the market cap is derived from speculative holding and long-term security (high SoV). A low ratio indicates that the majority of the market cap is supported by active utility and transactional usage (high MoE).

3 The Quadrant-Based Investment Framework

By plotting assets on a 2×2 matrix, centered on the median or average of each ratio within the asset class, we can derive investment conclusions:

Table 1: Quadrant-Based Investment Framework

	High SoV/MoE (Speculative Focus)	Low SoV/MoE (Utility Focus)
High NVT/G (Overvalued Utility)	AVOID High speculation; expensive MoE component.	MATURE / OVERPRICED High current utility; but MoE component is priced highly relative to growth. Potential value trap.
Low NVT/G (Undervalued Utility)	STAR / ECONOMIC MOAT High speculative holding; but the transactional (MoE) component is cheap and growing rapidly. Suggests a long-term economic moat.	GARP (Growth At Reasonable Price) Value is driven by utility; MoE component is priced reasonably relative to growth. Strong balance of value and utility.

This quadrant framework forces investors to look beyond simple market capitalization and price, analyzing the asset's functional split (SoV vs. MoE) against its value proposition relative to usage and growth (NVT/G). It guides capital towards assets that are cheap based on current or future utility, regardless of their current speculative hype.